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Roll No. _____

23/3105
B.C.A. (Fifth Semester)
Examination, 2023
First Paper
(Introduction to DBMS)

Time : 3:00 Hours / [Maximum Marks : 75

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) What are integrity constraints? Briefly discuss the different types of integrity constraints. 7½
(b) Explain the responsibilities of a database Administrator. 7½
2. Explain ANSI-SPARC 3 Level architecture of DBMS and type of data independence between different levels give the suitable diagram in support of your explanation. 15

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3. What is Relational Algebra? What is the utility of relational algebra? Is SQL related to relational algebra? Explain the following operations in relational algebra with the help of an example for each. 15

- (i) Select
- (ii) Project
- (iii) Join

4. (a) What do you understand by the term "Normalization" in DBMS? Write statement for second normal form (2NF) and discuss the insert, delete and update anomalies associated with 2NF, with the help of suitable example. 10
(b) Consider the relation R (A,B,C,D,E) and the set of functional dependencies. 5
 $F = [A \rightarrow D, \{A, B\} \rightarrow C, D \rightarrow E]$

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which of the following is a candidate
'key'?

- (i) {A}
- (ii) {A,B}
- (iii) {A,E}

5. What is E-R diagram? Draw any E-R diagram
which demonstrates the following : 15

- (i) Entity (strong, weak)
- (ii) Attribute
- (iii) Multi-valued attribute
- (iv) Composite attribute
- (v) Derived attribute

6. (a) Explain "ACID" properties of transaction
in detail. 8

(b) How does a checkpoint mechanism
help in database recovery? Explain
through an example. 7

7. What do you understand by Relational
model of database system? Elaborate the
major characteristics of relational Database
management system. 15

8. (a) Define Data manipulation language
(DML) of SQL. List and explain various
DML commands. 8

(b) What is the difference between
Authentication and Authorization. 7

9. Write about the following with suitable
example : 15

- (i) Hashing
- (ii) Candidate key
- (iii) Foreign key